



AgentConnect®

Introduction to AgentConnect

Version 15.12 (SP1)

2018

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The following diagram explains the formats used in process flows throughout this document.

For each API description, a red arrow points to the API being described.

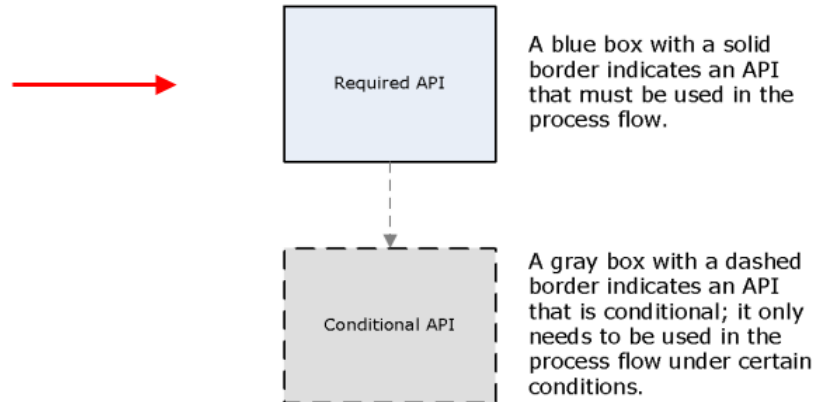


Figure 1: Explanation of Process Flow Diagrams

The following is an example of a table used to describe API request field details.

Tag	Datatype	Information and Rules	Input	Data
requestExampleField1	string Maximum length: 10	An example of a field containing a string.		
requestExampleField2	string Length: 3 Must be a valid country code	An example of a field containing a country code		
requestExampleField3	boolean	An example field whose input and data requirements depend on the <code>getFieldsForProduct</code> API. <i>Additional Information</i> 1. If <code>true</code> , then... 2. If <code>false</code> , then...		

Definitions

- **Tag:** The field's XML tag; fields for which data is required are shown in bold red text
- **Datatype:** Describes the type and restrictions of the field's data
- **Information and Rules:** Provides additional information, business rules, and a description of the field if necessary
- **Input:** Indicates how the field's value is obtained
 - **Application:** The agent application must retrieve the data from another part of the agent application or from a previous MoneyGram API response. Data can be displayed on the application screen
 - **GUI:** The agent application must visibility the field on its user interface and allow user input
 - **Conditional:** The agent application may or may not visibility the field on the user interface for user input depending on the conditions stated in the Information and Rules column
 - **GFFP:** Whether or not the field is shown on the user interface for user inputs depends on the information returned by the `getFieldsForProductResponse` (GFFP)
 - **N/A:** Input does not apply; for instance, if the field only contains other fields

- **Not Allowed:** The agent application must not allow data to be entered in the field by the user. Data must not be displayed. The agent application must retrieve the data from another part of the agent application or from a previous MoneyGram API response
- **Optional:** The agent may choose whether to visibility the field on the user interface for user input
- **Data:** Indicates whether the field must contain data; only used on API requests
 - **Required:** The field must contain data
 - **Not allowed:** The field must not contain data
 - **Optional:** The agent must display the field on the user interface, but data collection is optional
 - **Conditional:** The field may or may not contain data depending on the conditions stated in the Information and Rules column
 - **GFFP:** Whether or not the field contains data depends on the information returned by the `getFieldsForProductResponse` (GFFP)
 - **Response:** Whether or not the field contains data depends on the information returned by a message specified in the additional information for the API

The following is an example of a table used to describe API response field details.

Tag	Datatype	Information and Rules	Display
<code>responseExampleField1</code>	string Maximum length: 10	An example of a field containing a string.	required
<code>responseExampleField2</code>	string Length: 3 Must be a valid country code	An example of a field containing a country code	optional
<code>responseExampleField3</code>	boolean	An example field whose data and display requirements depend on the <code>getFieldsForProduct</code> API. <i>Additional Information</i> 1. If <code>true</code> , then... 2. If <code>false</code> , then...	required

Definitions

- **Tag:** The field's XML tag; fields for which data is required are shown in bold red text

- **Datatype:** Describes the type and restrictions of the field's data
- **Information and Rules:** Provides additional information, business rules, and a description of the field if necessary
- **Display:** Indicates whether the value of the field is shown on the agent application's user interface
 - **Conditional:** The value may or may not be shown on the user interface depending on the conditions stated in the Information and Rules column
 - **GFFP:** Whether or not the value is shown on the user interface depends on the information returned by the `getFieldsForProductResponse` (GFFP)
 - **N/A:** Display does not apply; for instance, if the field doesn't contain displayable data
 - **Not allowed:** The value must not be shown on the user interface; for fields in API responses, the data in the field should not be edited unless otherwise stated
 - **Optional:** The agent may choose whether to show the value on the user interface
 - **Required:** The value must be shown on the user interface if data is returned in the API

Introduction

This document provides an introduction to MoneyGram's AgentConnect® product, including descriptions of the technologies used, the product development cycle, and installing the AgentConnect simulator.

AgentConnect® provides APIs (application programming interfaces) to integrate MoneyGram money transfer products into your own POS (point-of-sale). There are four parts to the AgentConnect product:

1. A complete SDK (software development kit), including the detailed technical specification of the APIs and a simulator that allows developers, business analysts, and functional experts to see and to test messages.
2. MoneyGram consulting and advisory services to facilitate the design, development, and testing processes.
3. MoneyGram project management assistance.
4. MoneyGram quality assurance certification event.

In order to successfully develop a quality application, agents must have the following:

1. Familiarity with XML (Extensible Markup Language)
2. In-house or third-party systems development
3. Testing capabilities
4. A commitment to partner with MoneyGram to ensure mutual success

There are several benefits to using AgentConnect to perform wire transfer service with MoneyGram, including the following:

1. **Hardware and software independence:** Applications can be developed using any language and hardware platform as long as it can send and receive the appropriate MoneyGram Web server XML messages. This provides the ability to add MoneyGram wire transfer applications to existing POS or ATM systems.
2. **Integral transaction security:** Transaction security that is built into AgentConnect. All messages between agents and the MoneyGram Web server use HTTPS, so every message is encrypted. As an additional security measure, MoneyGram configures agents to use unique profiles, and either a password or token. By authenticating every transaction, regardless of the method of security chosen, MoneyGram ensures that two applications (the real agent's and an imposter's) cannot run using the same profile.
3. **Design flexibility:** Agents can create a user interface that is user friendly and appeals to consumers and employees. MoneyGram transactions can be integrated into existing applications, or agents can create specialized MoneyGram transaction applications. Agents control the "look and feel" of their wire transfer products.
4. **Assistance from MoneyGram:** MoneyGram has documented the AgentConnect commands (including standard messages, business rules, error messages and development tips). MoneyGram has also established a development environment available to outside developers for testing applications. In addition, AgentConnect includes a simulator that developers can use to see the correct format of outgoing and incoming messages, and send messages they create on their own.

Aside from offering tools for development, a dedicated MoneyGram team is ready to help. The team offers design advice, project management, and a certification event to ensure that agents can successfully process MoneyGram transactions.

Technologies

XML Schema and WSDL

Schema

AgentConnect uses XML messaging between the agent application and MoneyGram. Messages are described in an XML schema document which describes a set of elements, rules, and other information required to create valid AgentConnect messages. The XML schema document is written XML Schema, a [W3C](http://www.w3.org/) recommendation.

The following is an example of an element described by XML Schema.

```
<xs:element name="exampleParentElement" type="ac:ExampleParentElement"/>
<xs:complexType name="ac:ExampleParentElement">
  <xs:complexContent>
    <xs:extension base="ac:exampleElementBase">
      <xs:sequence>
        <xs:element name="exampleChildElement1" type="xs:string"/>
        <xs:element name="exampleChildElement2" type="xs:string"/>
        <xs:element name="exampleChildElement3" type="xs:string"/>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
```

The first line of code defines an XML element with the name `exampleParentElement`.

The second and third lines define the type of content contained element. In this case the element contains complex content; that is, it contains other elements.

The fourth line indicates the element extends another element; it contains the elements of `exampleElementBase` and adds further elements of its own.

The fifth line indicates that the element's child elements must occur in an XML document in the order they are shown in the schema.

The sixth, seventh, and eighth lines define the element's child elements. In this example, the elements' values are strings.

More information on XML Schema can be found online at the W3C's Web site, www.w3c.org.

WSDL

In addition to XML Schema, AgentConnect uses WSDL (Web Services Description Language), an XML language used to describe Web services. WSDL describes Web services as a set of endpoints operating on messages containing either document-oriented or procedure-oriented information.

MoneyGram supports SOAP version 1.2, WSDL version 1.1, and is compatible with WS-I Basic Profile 1.0. The message format conforms to literal schema instances, and the Web service uses document/literal as a binding style.

A complete description of WSDL can be found online at the W3C's Web site, www.w3c.org.

AgentConnect XML Schema

The XML Schema definition of an AgentConnect message appears as follows:

```
<xs:element name="exampleMessage" type="ac:ExampleMessage"/>
<xs:complexType name="ac:ExampleMessage">
  <xs:complexContent>
    <xs:extension base="ac:exampleMessageBase">
      <xs:sequence>
        <xs:element name="exampleMessageBodyField1" type="xs:string"/>
        <xs:element name="exampleMessageBodyField2" type="xs:string"/>
        <xs:element name="exampleMessageBodyField3" type="xs:string"/>
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
```

The schema defines the AgentConnect message as an element containing other elements. The child elements are of two types: header elements and body elements. The header elements are included in the message definition by extending the `exampleMessageBase` element. The body elements are defined explicitly.

If the header elements were defined explicitly, the entire message definition would appear as below:

```
<xs:element name="exampleMessage" type="ac:ExampleMessage"/>
<xs:complexType name="ac:ExampleMessage">
  <xs:complexContent>
    <xs:sequence>
      <xs:element name="exampleMessageHeaderField1" type="xs:string"/>
      <xs:element name="exampleMessageHeaderField2" type="xs:string"/>
      <xs:element name="exampleMessageHeaderField3" type="xs:string"/>
      <xs:element name="exampleMessageHeaderField4" type="xs:string"/>
      <xs:element name="exampleMessageBodyField1" type="xs:string"/>
      <xs:element name="exampleMessageBodyField2" type="xs:string"/>
      <xs:element name="exampleMessageBodyField3" type="xs:string"/>
    </xs:sequence>
  </xs:complexContent>
</xs:complexType>
```

More information on the types of extensions used in AgentConnect messages is found in [Message Headers](#).

AgentConnect Messages

An Agent application must have the capability to capture all request and response xml messages with MoneyGram.

Communication between the agent application and MoneyGram consists of a request message sent from the agent application to MoneyGram, followed by a response message from MoneyGram to the agent application. All messages to and from AgentConnect use HTTPS, which is HTTP (Hypertext Transfer Protocol) combined with TLS (Transport Layer Security). The agent application opens a network connection to MoneyGram and sends HTTP headers according to the HTTP protocol and a SOAP data stream for the AgentConnect API on Port 443.

Format

An AgentConnect message consists of a set of standard headers combined with an element defined by the AgentConnect XML schema.

1. HTTP header
2. XML declaration
3. SOAP envelope
4. Message name
5. AgentConnect message header fields
6. AgentConnect message body fields

A complete AgentConnect message appears as in the example below.

```
POST http://example.com/servicesLocation HTTP/1.1
User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; MS Web Services Client
Protocol 4.0.30319.225)
VsDebuggerCausalityData:
uIDPolWZA/LuXitCvkmnX3Hc9b0AAAAA4uIDS8IVgUe3j6BgwkHnzPT5YTeFuklPrhZDCTTFXlIAC
QAA
Content-Type: text/xml; charset=utf-8
SOAPAction: "urn:example#exampleMessageName"
Host: example.com
Content-Length: 000
Expect: 100-continue
<?xml version="1.0" encoding="utf-8"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <soap:Body>
    <exampleMessageName xmlns="http://example.com/schemaLocation">
      <exampleMessageHeaderField1>Header Field 1
Value</exampleMessageHeaderField1>
      <exampleMessageHeaderField2>Header Field 2
Value</exampleMessageHeaderField2>
      <exampleMessageHeaderField3>Header Field 3
Value</exampleMessageHeaderField3>
      <exampleMessageBodyField1>Header Field 1 Value</exampleMessageBodyField1>
      <exampleMessageBodyField2>Header Field 2 Value</exampleMessageBodyField2>
      <exampleMessageBodyField3>Header Field 3 Value</exampleMessageBodyField3>
    </exampleMessageName>
  </soap:Body>
</soap:Envelope>
```

SOAP

MoneyGram implements SOAP version 1.2 using Apache SOAP version 1.1.

It may be easier for the developer to use SOAP to generate messages to send to the Host. However, SOAP is not required to use the AgentConnect API, and it is not documented here.

Message Headers

AgentConnect request and response messages use the following common header fields:

Tag	Datatype	Information and Rules	Display
unitProfileID	int	Internal Use Only.	Not allowed
agentID	string Length: 8	The agent's unique identification number. Each location will have a unique agent ID number assigned by MoneyGram which must be used on every API call.	Not allowed
locationCode	string Length: 8	1.A unique ID assigned to a particular Point of Sale. 2. Not allowed when an agentID or unitProfileID is used.	Not allowed
agentSequence	string Maximum length: 2	The unique POS number that identifies an individual POS at the agent location. Other locations may have POS devices with the same number. The number is provided by MoneyGram and should not be hard-coded by the agent.	Not allowed
token	string Maximum length: 20	The password for the particular Agent POS.	Not allowed

language	string	The language in which errors and other language-specific information is returned. <i>Additional Information – the following languages are supported, but not all error messages are returned in all the languages listed. If translation is not available, English will be returned.</i>	
	Default value: en		

Tag	Datatype	Information and Rules	Display
timeStamp	dateTime Format: CCYY-MM-DDThh:mm:ss.sss-hh:mm	The date and time the request was sent.	
apiVersion	string Valid values: 1512	The API version the agent application uses.	
clientSoftwareVersion	string Maximum length: 6	The version of the agent application being used. This value is set by the agent and can be used for debugging purposes. MoneyGram stores this information in its systems, but does not check or return the value. It is used when an agent places a technical call to verify problems between different versions.	
userID	string Maximum length: 20	Internal Use Only.	

Tag	Datatype	Information and Rules	Display																		
channelType	string	Identifies the type of channel for the POS application																			
		<table><tr><th>Channel Type Code</th><th>Description</th></tr><tr><td>ATM</td><td>Transaction is done using ATM that has MoneyGram products enabled.</td></tr><tr><td>KIOSK</td><td>Transaction is done with either a MoneyGram or agent owned kiosk.</td></tr><tr><td>LOCATION</td><td>Transaction is done at a physical location (Brick & Mortar)</td></tr><tr><td>MOBILE</td><td>Transaction is done using either a MoneyGram owned or agent owned web interface via a mobile device</td></tr><tr><td>SYSTEM</td><td>Transaction is done using backend processes or rules.</td></tr><tr><td>WEB</td><td>Transaction is done using either a MoneyGram owned or agent owned web interface. Does not include transactions through a mobile device.</td></tr><tr><td>TABLET</td><td>Transaction staged using Tablet device either a MoneyGram owned or agent owned web interface. Does not include transactions through a Kiosk or Mobile device.</td></tr><tr><td>OTHER</td><td>Other staging initiation and completion Channel</td></tr></table>		Channel Type Code	Description	ATM	Transaction is done using ATM that has MoneyGram products enabled.	KIOSK	Transaction is done with either a MoneyGram or agent owned kiosk.	LOCATION	Transaction is done at a physical location (Brick & Mortar)	MOBILE	Transaction is done using either a MoneyGram owned or agent owned web interface via a mobile device	SYSTEM	Transaction is done using backend processes or rules.	WEB	Transaction is done using either a MoneyGram owned or agent owned web interface. Does not include transactions through a mobile device.	TABLET	Transaction staged using Tablet device either a MoneyGram owned or agent owned web interface. Does not include transactions through a Kiosk or Mobile device.	OTHER	Other staging initiation and completion Channel
		Channel Type Code		Description																	
		ATM		Transaction is done using ATM that has MoneyGram products enabled.																	
		KIOSK		Transaction is done with either a MoneyGram or agent owned kiosk.																	
		LOCATION		Transaction is done at a physical location (Brick & Mortar)																	
		MOBILE		Transaction is done using either a MoneyGram owned or agent owned web interface via a mobile device																	
		SYSTEM		Transaction is done using backend processes or rules.																	
		WEB		Transaction is done using either a MoneyGram owned or agent owned web interface. Does not include transactions through a mobile device.																	
		TABLET		Transaction staged using Tablet device either a MoneyGram owned or agent owned web interface. Does not include transactions through a Kiosk or Mobile device.																	
OTHER	Other staging initiation and completion Channel																				

Tag	Datatype	Information and Rules	Display
targetAudience	string valid values: AGENT_FACING CONSUMER_FACING	Identifies if the application is agent facing or consumer facing. The fields labels returned in GetFieldsForProduct (GFFP) and error messages can be different based on this field. They labels and messages will be directed toward agent or consumer	

Tag	Datatype	Information and Rules	Display
doCheckIn	boolean	<i>Additional Information</i> 1. Field is not currently used.	Not allowed
timeStamp	dateTime Format: CCYY-MM-DDThh:mm:ss.sss-hh:mm	The date and time the request was sent from agent application.	
Flags	int	If agent application is performing regular calls on the maintenance APIs, this value can be ignored These bit set values added together can produce combinations of information that should be downloaded. For example, a 3 returned indicates that the agent application should retrieve a new code table and profile information. Since this is a bit set, applications need not consider flags that are not relevant to it. For example, if the application receives an 11 ($1 + 2 + 8 = 11$), it only has to consider the 1 and 2, and the 8 can be ignored. If the application receives a 6 ($2 + 4 = 6$), it only has to consider the 2, and the 4 can be ignored because it is deprecated. <i>Additional Information</i> 1 is new code table information is available 2 is new profile information is available 4 is not used 8 is not used 16 new translation information is available	

Error Messages

SOAP faults are returned whenever the Host cannot properly process a request. They normally include the source of the problem (server or agent), the short name of the problem, an error code, a detailed description of the problem, and other information.

Tag	Datatype	Information and Rules	Display
faultCode	string	Indicates where the problem originated. This can be client, server, or SOAP version mismatch.	
faultString	string	A short description of the error that occurred.	
detail	complex	Returned when the fault was generated by the AgentConnect server.	
errors	complex Maximum number of occurrences: unbounded	None	
errorCode	string	The error code specific to the problem that occurred.	Required
errorString	string maximum length: 500	A short description of the error that occurred.	Required
subErrorCode	string	The particular sub error code, if any, for the fault code specified above.	Required
offendingField	string	The request field that caused the error. This field appears only where valid.	optional
timeStamp	dateTime Format: CCYY-MM-DDThh:mm:ss.sss-hh:mm	The date and time the message was sent from MoneyGram server.	
detailString	string maximum length: 500	More detailed information on what may have caused the error.	Required

The agent application must display the following on screen:

- `errorCode`
- `errorString`
- `subErrorCode`
- `detailString`

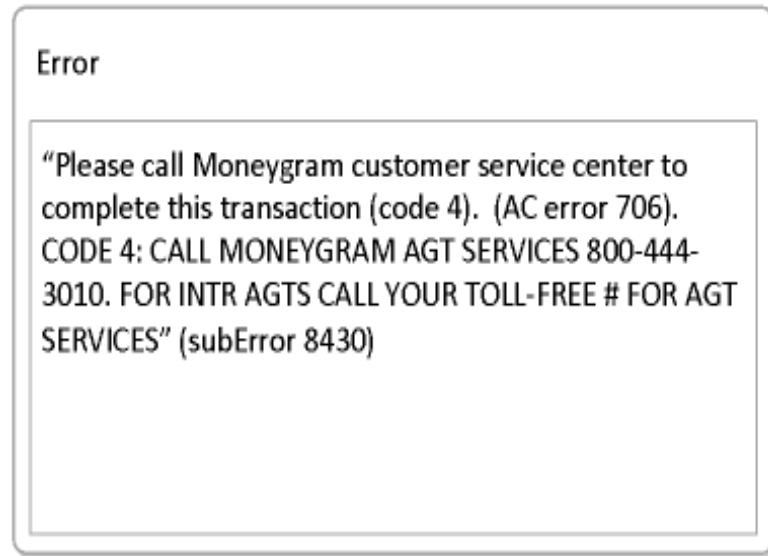


Figure 2: Example Error Dialog

When a transaction is submitted, MoneyGram will perform validations on the transaction and return any and all errors that are found for the transaction. For example, if there are 2 required fields were not submitted and another field that contains invalid data, MoneyGram will return an array of 3 different error messages indicating what is incorrect with the transaction. The application must be able to handle mutiple error messages, correct all errors before re-submitting the transaction to MoneyGram again.

International Character Support

MoneyGram supports the following international characters. These accented characters are converted to non-accented characters in MoneyGram's systems.

From	To
À, Á, Â, Ã, Ä, Å	A
È, É, Ê, Ë	E
Ì, Í, Î, Ï, I	I
Ò, Ó, Ô, Õ, Ö, Ø	O
Ù, Ú, Û, Ü	U
Ý, Ÿ	Y
Ñ	N
Ç	C
Ğ	G
Ş	S
à, á, â, ã, ä, å	a
è, é, ê, ë	e
ì, í, î, ï, i	i
ò, ó, ô, õ, ö, ø	o
ù, ú, û, ü	u
ý, ÿ	y
ñ	n
ç	c
ğ	g
ş	s

Table 1: Character Mapping

High Network Availability

MoneyGram provides high network availability to Agents. MoneyGram currently has redundant datacenters, each with separate OLTP (online transaction processing environments) including redundant firewalls, load-balancers, web servers, and websphere application servers. Processing transactions at both datacenters significantly increases availability of MoneyGram services.

High Availability with Internet

The agent may or may not have redundant servers and uses Internet DNS to resolve xxx.moneygram.com. The agent does not translate MoneyGram IP addresses to other IP addresses.

The agent connects to the MoneyGram production environment via the Internet and uses public DNS servers for resolution. The agent must connect to the MoneyGram URL since that URL resolves to two different IP addresses located in different data centers. MoneyGram monitors the URL xxx.moneygram.com in each MoneyGram datacenter and makes dynamic changes to the Internet DNS. When resolving xxx.moneygram.com, the agent receives either the IP address 63.91.129.XXX or 207.67.74.XXX.

AgentConnect High Availability Diagram

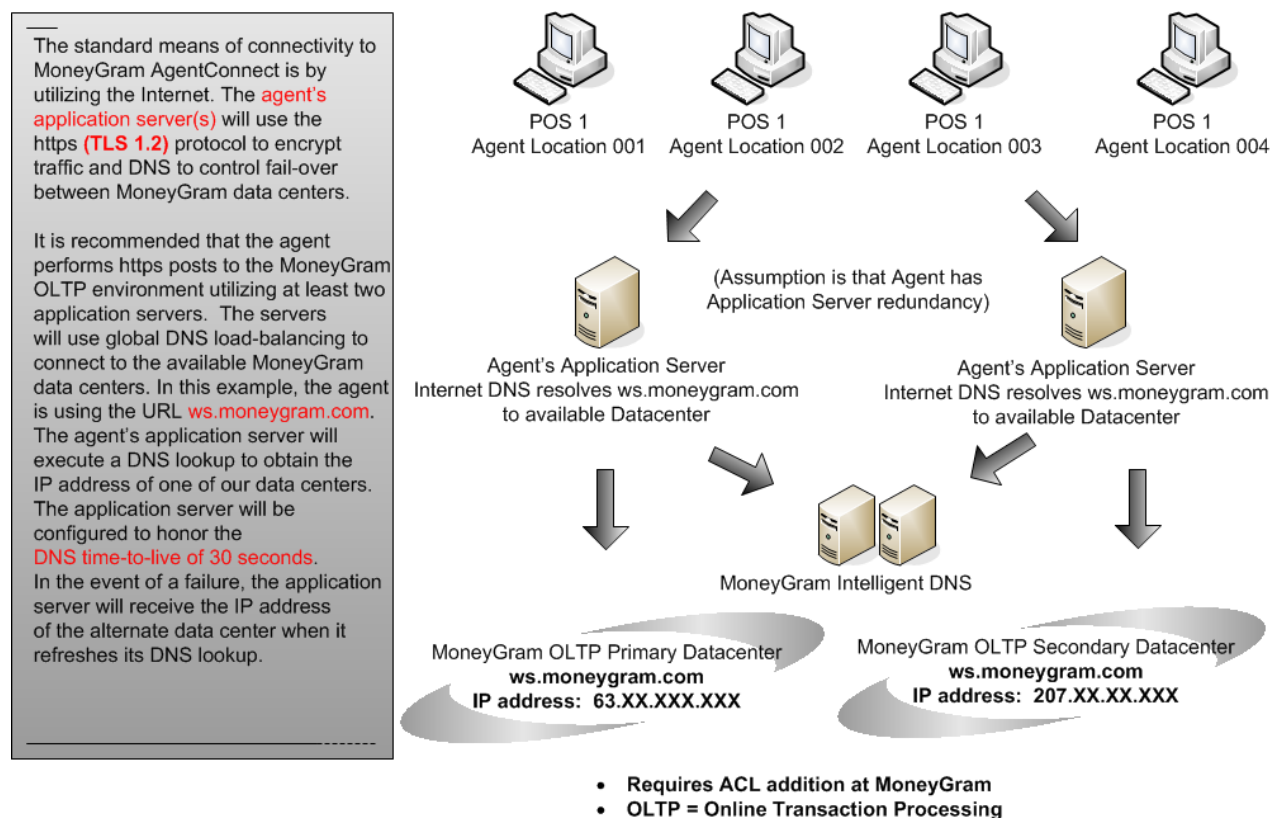
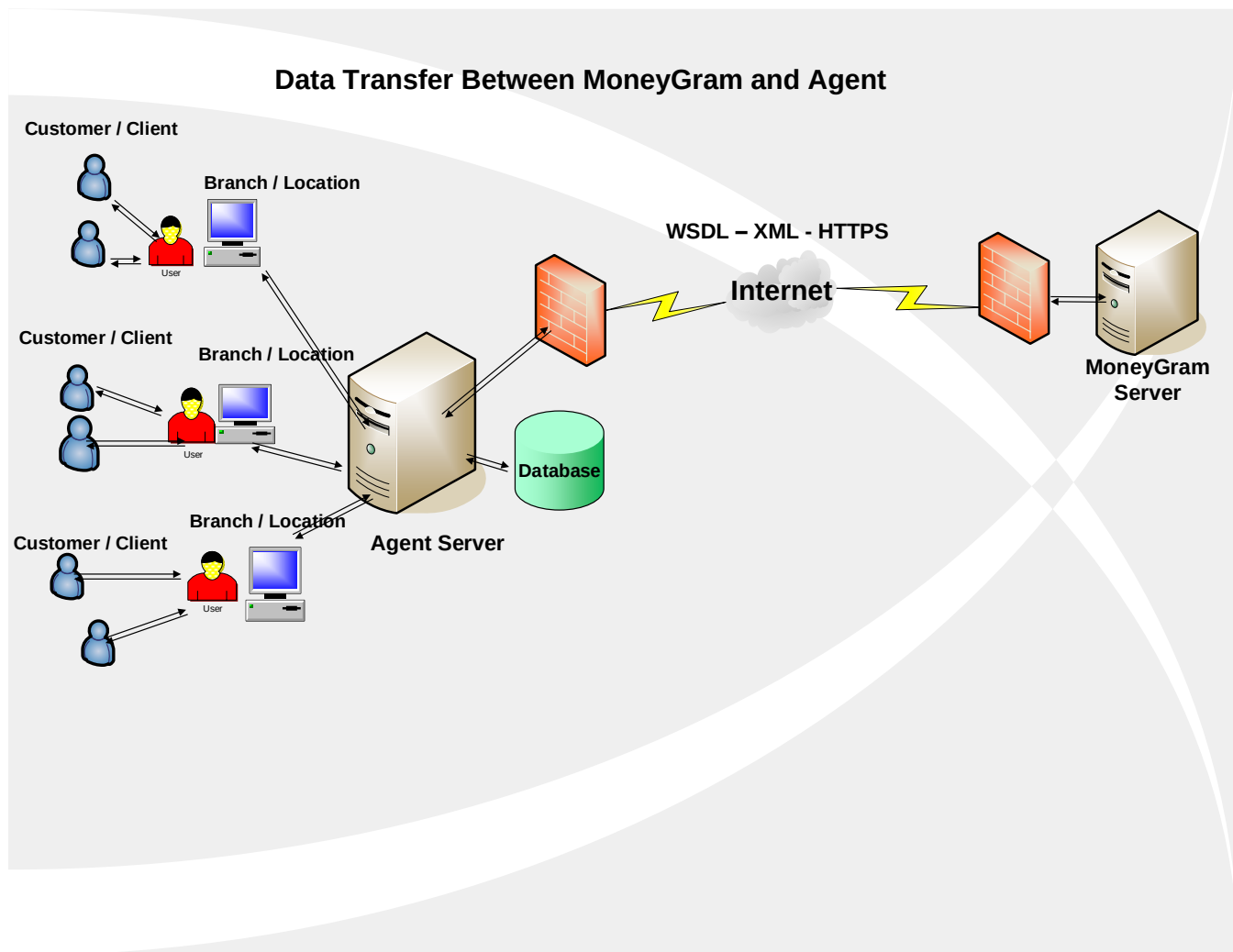


Figure 3: AgentConnect High Availability Diagram

Location and Agent ID setup



- MoneyGram assigns a unique agent ID for each location/branch.
- Agent ID, Agent sequence [POS#] and Token [Password] are required in the API message header, which are required for every API call to process the transaction.
- The production live credentials are provided by MoneyGram after the certification sign-off.
- The live credentials must be mapped on the agent side, so that the agent application can provide the correct assigned agent ID for the specified location

Example of setup

Location #1: AgentID 1234, POS/agentSequence #1, Password/token xxxxx

Location #2: AgentID 3849, POS/agentSequence #3, Password/token xxxxx

Location #3: AgentID 3546, POS/agentSequence #3, Password/token xxxxx

Location #4: Agent ID 1355, POS/agentSequence #2, Password/token xxxxx

Product Development Cycle

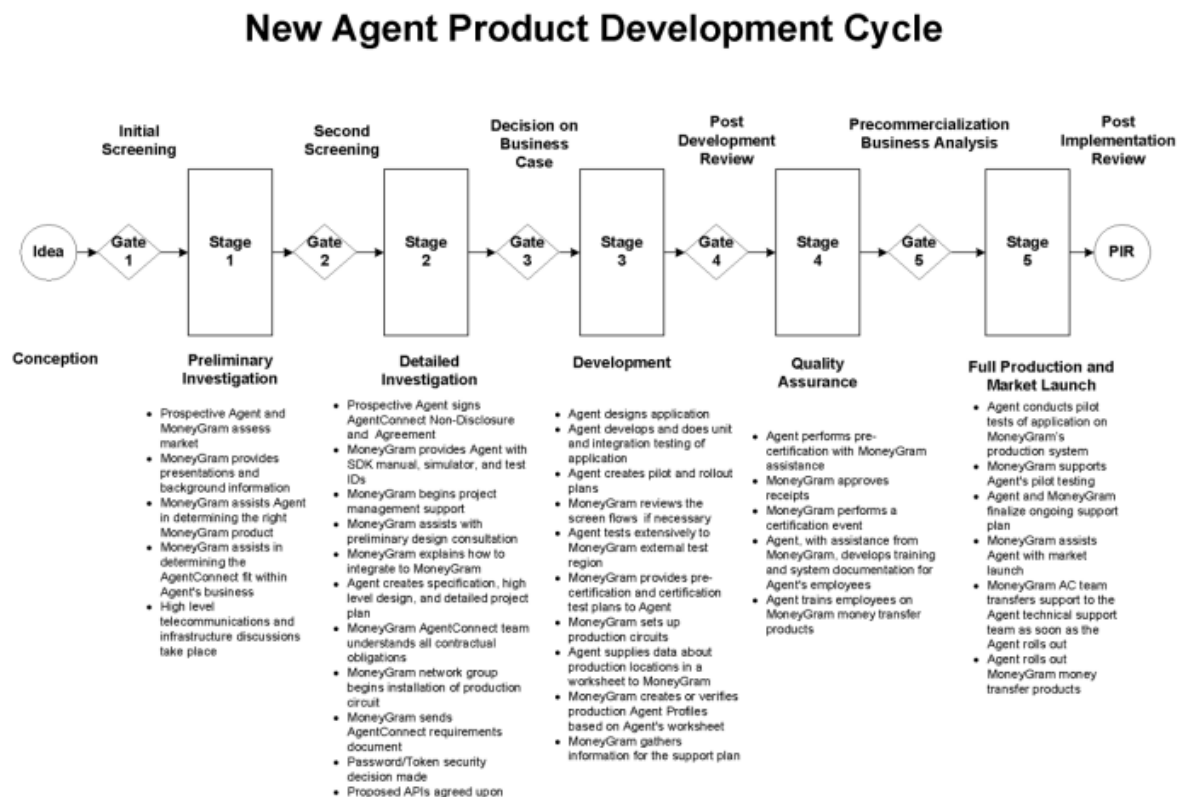


Figure 4: Product Development Cycle

The product development life cycle consists of the following steps:

1. Preliminary investigation
2. Detailed investigation
3. Development
4. Quality assurance
5. Full production and market launch

The agent's responsibilities during the five stages are described in the following sections. Understanding who performs which responsibilities, and what tasks occur in the sequence, helps create better expectations for product completion.

Preliminary Investigation

1. Agent and MoneyGram assess market
2. MoneyGram provides presentations and background information
3. MoneyGram assists agent in determining products
4. MoneyGram assists in determining fit of AgentConnect in agent's business
5. High level telecommunication and infrastructure communications take place

Detailed Investigation

1. Agent signs AgentConnect NDA and Agreement
2. MoneyGram provides agent with AgentConnect SDK materials
3. MoneyGram begins project management support
4. MoneyGram assists with preliminary design consultation
5. MoneyGram explains integration with MoneyGram systems
6. Agent creates specification, high level design, and project plan
7. AgentConnect team understands all contractual obligations
8. MoneyGram sends Agent Requirements form
9. Proposed APIs agreed upon

Development

1. Agent designs application
2. Agent creates and performs unit and integration testing of application
3. Agent creates pilot and rollout plans
4. Agent tests application in MoneyGram external region using test IDs provided by MoneyGram
5. MoneyGram provides pre-certification and certification test plans
6. Agent provides information about agent locations
7. MoneyGram creates or verifies agent profiles
8. MoneyGram gathers information for support plan
9. MoneyGram provides technical support and has regular calls with agent to discuss the APIs, business rules, questions, and other issues agent may encounter

Quality Assurance

1. Agent performs pre-certification with MoneyGram assistance
 - a. MoneyGram provides test cases. Agent executes test cases and provide xml results and screen shots. MoneyGram validate results.
2. MoneyGram approves receipts
3. Agent performs certification event with MoneyGram assistance. Same process as pre-certification in 1a.
4. Agent creates training and system documentation with MoneyGram assistance
5. Agent trains employees

Full Production and Market Launch

1. Agent performs pilot test
2. MoneyGram supports pilot testing
3. Agent and MoneyGram finalize ongoing support plan
4. MoneyGram assists agent with market launch
5. Agent rolls out MoneyGram products

Some sample test cases sent to Agent during pre-certification are given below. There would be similar test cases for other APIs and process flows the Agent would be developing. When test

cases are executed, agents need to provide complete xml logs of all test cases and screen shots in some cases.

MoneyGram Successful Send/Receive Test Cases						
Test Case Name	From	To	Instructions	Amount (in sending agent's currency)	Reference Number	Agent Comments
SN08	USA	USA	Perform a send transaction providing data in ALL fields using the following information on the initial send: - Receiver's First name: Pharringalbertastien - Receiver's Last name: Coomaraswamanushtanu - Receiver's Second Last name: ELcbertastienpherringcoladaren Please provide us the screen shot of complete flow and also provide screen shot of the receipt.	567.89		
RC07	SPAIN	UK	Perform a receive transaction providing data in only REQUIRED fields.	1500.00	78945621	
MoneyGram Unsuccessful Send/ Receive Test Cases						
			REMINDER: As with other negative test cases, please execute each of the following as far as you are able, using your un-altered client application. Following the attempt, provide all XML generated for the transaction (as with other cases); also, include the reason for your stopping point in the "comments" column to the right.			
SN40	USA	MEXICO	Perform a send providing data in only REQUIRED fields in your application. 1. Use one of the following for the sender's country: - United States - Mexico - Canada 2. DO NOT enter a value for the sender's State. A. Please return all XML generated. B. If not possible to execute the transaction or an error occurs, please return a comment describing your application's response. C. Return a screen shot showing how your application prevents this transaction from being completed, including any error messages that are displayed on your application's UI.	342.90		
RC21	USA	CHINA	Perform a Reference Number Lookup using the reference number given. A. Please return all XML generated. B. If not possible to execute the transaction or an error occurs, please return a comment describing your application's response. C. Return a screen shot showing how your application prevents this transaction from being completed, including any error messages that are displayed on your application's UI. Expected: The reference number is invalid and should return an error in the referenceNumber response.	N/A	11122236	

Submitting Problems to MoneyGram During Development

Information that MoneyGram needs in order to help developers solve issues is shown below. MoneyGram requires Agent developers to provide XML code with request and response information. The XML code is used during application certification stages and is also used if production issues arise in the future.

If issues are discovered during the development process, please submit the information that is requested below along with the issue to ensure a timely response. The information should be submitted to all appropriate technical and business persons.

After completion of development, there are different procedures to follow when the application is in production. These procedures are found in the support plan developed with MoneyGram.

- The date the issue occurred
- The name of the person who can be contacted with any questions on the issues and may be able to help in the resolution of the issue
- A brief descriptive title of the issue
- A thorough description of the issue
- Whether or not the issue is repeatable
- The steps taken to cause the issue
- The results of attempting to cause the issue using the AgentConnect Simulator

Pilot and Support Plans

Pilot Plan

The pilot plan provides an opportunity to perform a controlled release of AgentConnect to the production environment. It verifies successful interaction of AgentConnect messaging between the agent application and the MoneyGram system.

During the pilot phase, the agent application should be rolled out to a minimum of five locations. MoneyGram's preference is for an agent to have no more than ten locations to minimize production impacts if problems arise. The MoneyGram project team supports the agent during the entire pilot process and supports any issues that may arise during the pilot phase.

Support Plan

The purpose of the support plan is to provide a service guide to the agent and MoneyGram when the agent moves to full production following its pilot. The support plan is created by the agent and MoneyGram.

Installing the AgentConnect Simulator

The AgentConnect simulator is an application that creates and sends requests to MoneyGram and displays the responses it receives. For transactions that require more than one message, the simulator repeats this process for each subsequent step. The AgentConnect simulator can also be used to transmit AgentConnect messages that have been created externally (e.g., by hand, application or other method).

The simulator logs every message it sends or receives so that the syntax, structure, and fields can be reviewed at a later time.

Follow the following steps to install the AgentConnect simulator:

1. Execute `setup.exe`.
2. The InstallShield program will guide you through the installation process. The following directories and files are installed:
 - **Lib:** Contains the Java jar files needed for the simulator to run
 - **Etc:** Contains properties and settings files, including the `AgentConnectSimulatorSettings.xml` file
 - **Source:** Contains a .zip file of the Java source code for the simulator
 - **jre6:** Contains version 6 of the Sun Java Runtime Environment
 - **AgentConnectSimulator.bat:** The .bat file that sets up the Java classpath and runs the simulator
3. New files are created the first time the simulator is run:
 - **AgentConnectSimulatorSettings.xml:** Contains URL, WSDL location, Agent ID, and header information so the agent doesn't need to load this information every time the Simulator is started
 - **AgentConnectSimulator.log:** Logs all request and response messages in XML
4. For instructions on how to use the simulator, please refer to AgentConnect Simulator User Guide

Sample Test ID letter and Instruction to use the Simulator:

The AgentConnect simulator is a Java application designed for and tested on Windows Server 2008 R2, Windows Server 2008, Windows Server 2003, Windows 7, Windows Vista, Windows XP, Windows 2000 and all other higher versions of Windows as a development tool to assist you as you program your MoneyGram money transfer application.

Getting Started:

1. Install the simulator by running *setup.exe*.
The AgentConnect Simulator also allows agents to connect to MoneyGram using a proxy server. To connect to a proxy, open the simulator and click <Security> on the menu bar, then select <Proxy Settings> to input and save your settings. Be sure to check the Use Proxy checkbox.
2. Make sure it is possible to connect to the endpoint URL.
This is used to send requests to the AgentConnect server from the simulator and the Agent's application. <https://extws.moneygram.com/extws/services/AgentConnect1512>
This can be run in a browser to verify connections with the AgentConnect server.
3. Point the simulator at the WSDL file.
The WSDL is used by the simulator to assist in formatting XML
<https://extws.moneygram.com/extws/AgentConnectWSDL?Version=1512>
4. Run sample transactions using your test agent IDs.
Listed below are *few* samples of MoneyGram Agent identification numbers and their associated POS numbers and passwords for you to use during testing. Use one of these for "send" transactions and the other for "receive" transactions. It does not matter which ID is used for the "send" or the "receive" as long as "send" and "receive" are not used from the same ID on one transaction. MoneyGram has a business rule that prevents the same Agent from sending and receiving the same transaction.
 - Test ID 1
Agent ID: xxxxxxxx
POS Number: xx
Password: TEST
 - Test ID 2
Agent ID: xxxxxxxx
POS Number: xx
Password: TEST

If your test IDs become deauthorized during application development, you can reauthorized them as follows:

The re-authorization URL works by running it from a browser command line. Re-authorize a de-authorized development POS ID at

<https://extws2.moneygram.com/extwsweb/servlet/ReauthServlet>

Please note that this method of re-authorization is only allowed in the test environment. Once an Agent begins running production transactions, the Agent must call MoneyGram's Help Desk to re-authorize a POS device. Contact information for re-authorization will be distributed to the Agent prior to production rollout.

Additional information on Proxy configuration:

If Proxy configuration is needed to allow the AgentConnect Simulator to access the internet, the following Proxy Settings Information must be configured:

Security>Proxy Settings

The image shows a 'Proxy Settings Information' dialog box with the following fields and controls:

- Use Proxy :** A checkbox that is checked.
- Proxy Type:** A dropdown menu currently showing 'HTTPS'.
- Proxy Host:** A text input field.
- Proxy Port:** A text input field.
- User Id:** A text input field.
- Password:** A text input field.
- Buttons:** 'Save', 'Clear', and 'Cancel' at the bottom.

Red arrows point from a table of annotations to the corresponding fields in the dialog box:

Proxy Host: Proxy Server Name or IP address
Proxy Port: 443
User Id: User ID to authenticate with Proxy Server (If applicable)
Password: Password to authenticate with Proxy Server (If applicable)

Figure 5: Proxy Configuration Screen

PCI Compliance

MoneyGram recognizes and values the importance of enacting information security best practices throughout our organization. In order to provide our agents with a secure network for communicating and doing business with us, we recommend that business partners and agents who connect directly to our network follow the Payment Card Industry (PCI) security standards. These security standards have been adopted as security best practices among the financial and payment services industries.

More information may be found at the PCI Council website: <https://www.pcisecuritystandards.org/>

Initialization and Maintenance

Business Purpose

1. This section provides technical and business information required to initialize a POS and keep an agent application up to date with critical system information.
2. The agent must store two types of system information for use in APIs.
 - a. **Profile information:** Contains agent application system settings such as available products, financial limits, security settings, and network settings
 - b. **Code table information:** Contains MoneyGram system information such as country, state/province, and currency codes used by MoneyGram, available transaction delivery options, and rewards program information.

API Flows and Processes

POS Initialization Process

The agent application sends a non-financial transaction API containing the password provided by MoneyGram in the message header, and MoneyGram returns the corresponding response. Examples of non-transactional APIs: `feeLookup`, `referenceNumber`, `codeTable`

Maintenance Process

1. Choose one of the following.
 - a. The agent application sends a [codeTableRequest](#), and MoneyGram returns a [codeTableResponse](#) containing country, currency, delivery option, and other information that must be used by the application.
 - b. Information that must be used by the application is obtained by using a combination of the following:
 - The agent application sends a [countryInfoRequest](#), and MoneyGram returns a [countryInfoResponse](#) containing country information.
 - The agent application sends a [stateProvinceInfoRequest](#), and MoneyGram returns a [stateProvinceInfoResponse](#) containing state and province information. Currently the list of states/provinces are returned for Canada, Mexico, and the United States. However, the countries that have states/provinces available will increase and agent's application must be able to handle these.
 - The agent application sends a [currencyInfoRequest](#), and MoneyGram returns a [currencyInfoResponse](#) containing currency information.
2. For US Send agents only: `doddFrankStateRegulatorInfo` API is required to obtain information for receipts
3. [Optional] The agent application sends a [translationsRequest](#) containing a language code, and MoneyGram returns a [translationsResponse](#) containing some code table information translated into the requested language.
4. The agent application stores the returned information.

POS Profile Process

1. The agent application sends a [profileRequest](#) with message header information for the POS for which to get information, and MoneyGram returns a [profileResponse](#) containing POS profile information.
2. The agent application stores the returned information.

APIs

Required APIs

1. codeTable

Conditional APIs

2. profile
3. countryInfo
4. currencyInfo
5. stateProvinceInfo
6. translations
7. doddFrankStateRegulatorInfo
8. variableReceiptInfo
9. industry

1. *profile*

The agent application uses the [profile](#) API to obtain information used to comply with business rules when performing transactions. Information includes limits and permissions about how the agent profile is setup.

Additional Information

1. If an agent supports third party, or is located in the US, the third limit is returned in this API. Agents must store information returned in the [profileResponse](#).
2. The agent must create an automated process to obtain updates during non-business hours; the agent must also create a manual process to obtain updates in the event profile changes occur during business hours
3. The agent profile only needs to be updated by one location using the model agent ID; the information may be shared among locations

profileRequest

Field Summary

- [\(None\)](#)

Field Detail

Tag	Datatype	Information and Rules	Input	Data
(None)	N/A	None	N/A	

profileResponse

Field Summary

- profileItem
 - index
 - key
 - value
- productProfileItem
 - index
 - key
 - value
 - productID

Additional details about some of the important profile and product profile items:

1. key: AGENT_TIME_ZONE: Agent's Time zone information as setup in MGI system (example value: GMT+1:00)
2. key: AGENT_TIME_ZONE_NAME (example value: Europe/Paris)
3. key: AGENT_DST (values: Y or N). Y indicates Daylight saving applied)
4. key: AGENT_HOURS (example value: 09:30:00,18:30:00 represents opening and closing hours respectively. Index: 1 to 7 represents Sun to Sat)
5. key: ACTIVATE_REWARDS_CARD_OPTION (example value: N): indicates whether the Plus enrollment is possible by this Agent ID.
6. key: PROMOTIONAL_CODE_LIMIT (value: 0, 1 or 2): indicates the maximum number of promotional codes allowed to be used during a Send transaction.
7. key: AGENT_NAME (Agent's DBA name)
8. key: AGENT_ADDRESS_1, AGENT_ADDRESS_2, AGENT_ADDRESS_3, AGENT_CITY, AGENT_STATE, AGENT_ZIP : returns the detailed address information as in MGI system.
9. key: AGENT_CURRENCY - Send currency information, if the Agent is multicurrency Send, this will be returned multiple times. The index indicates the currency code, and the value indicates the 3-letter currency code.
10. key: ALLOW_LANG_OPT (valid values: eng, spa, fra, deu, zho) : indicates receipt languages for primary and secondary receipts.
11. key: ALLOW_NO_SECONDARY_LANG, (valid value: N or Y): if N indicates no need have the receipt in a secondary language
12. key: HDV_LIMIT , the high dollar verification limit. Currently not applicable to AgentConnect.
13. key: RECEIVE_CURRENCY: values could be 3-letter currency codes. if the Agent is multicurrency Receive active, this could be returned multiple times
14. key: USE_OLD_NAME_SIZES (value: Y or N)
15. key: DOCUMENT_NAME (example value: MONEYGRAM SEND, productID: 5): indicates ID and name for a particular product
16. key: FRAUD_LIMIT_LEGAL_ID - indicates the
17. legal Id verification threshold for the particular product and currency.
18. key: FRAUD_LIMIT_THIRD_PARTY_ID - indicates the third party ID verification threshold for the particular product and currency.
19. key: FRAUD_LIMIT_PHOTO_ID - indicates the photo Id verification threshold for Send and particular currency.
20. key: FRAUD_LIMIT_TEST_QUESTION - indicates the test question and answer threshold for Send and particular currency.
21. key: ALLOW_DEL_OPT: indicates whether transaction in a particular delivery option may be attempted. If value:N for the particular delivery option ID that would mean it is not allowed.
22. key: FRAUD_LIMIT_CASH_WARNING - not applicable to AgentConnect
23. key: MAX_AMOUNT_PER_ITEM - indicates the maximum amount per transaction for particular product and currency.
24. key: FRAUD_LIMIT_ADDRESS - indicates the receiver address verification limit for Receive and particular currency
25. key: DEFAULT_CASH_PAYOUT_AMOUNT - not applicable to AgentConnect
26. key: FRAUD_LIMIT_RECEIVER - indicates the photo Id verification threshold for Receives and particular currency
27. key: PRODUCT_STATUS (example value: A) - indicates whether a particular product is available. For example, if value is A for product ID: 6 it would mean that this Agent ID is authorized to perform Receive transactions.
28. Key: FRAUD_TURN_OFF_DOB_ENABLE
29. Key: FRAUD_TURN_OFF_COB_ENABLE
30. Key: FRAUD_LIMIT_COB
31. Key: FRAUD_LIMIT_DOB
32. Key: FRAUD_LIMIT_PERSONAL_ID_2
33. Key: FRAUD_LIMIT_THIRD_PARTY_ID
34. Key: DAILY_BILL_PAY_LIMIT
35. Key: FORM_FREE_ENABLED

Field Detail

Tag	Datatype	Information and Rules	Display
profileItem	complex Maximum number of occurrences: unbounded	None	N/A
index	int	The order of the profile item. <i>Additional Information</i> 1. Value is 0 unless multiple profile items use the same value for the <code>key</code> field; for example, store hours are indexed 1-7 for days of the week	N/A
key	string Maximum length: 40	The name of the profile item.	N/A
value	String Maximum length: 400	The value of the profile item	N/A
productProfileItem	complex Maximum number of occurrences: unbounded	None	N/A
index	int	The order of the product profile item. <i>Additional Information</i> 1. Value is 0 unless multiple profile items use the same value for the <code>key</code> field; for example, store hours are indexed 1-7 for days of the week 2. For multi-currency send or receive transaction, the value refers to the specific currency; for example index of 840=USD	N/A

Tag	Datatype	Information and Rules	Display
key	string Maximum length: 40	The name of the product profile item.	N/A
value	string Maximum length: 400	The value of the product profile item	N/A
productID	int	The numeric ID of the product profile item <i>Additional Information</i> 1. 5 is MoneyGram Send 2. 6 is MoneyGram Receive 3. 7 is Bill Payment 4. 12 is Receiver Registration	N/A

2. *codeTable*

The agent application uses the [codeTable](#) API to obtain countries, states, currencies, delivery options, and other important information required to perform transactions.

Additional Information

1. Agents must store the information returned in the [codeTableResponse](#) locally
2. The agent must create an automated process to obtain updates during non-business hours; the agent must also create a manual process to obtain updates in the event system changes occur during business hours
3. The code table information only needs to be updated by one location using the model agent ID; the information may be shared among locations
4. This process should be performed on a regular basis. MoneyGram recommends a daily batch process during non-business hours.
5. In the codeTableResponse, the agent can keep track of MoneyGram's 'version' in the agent's local table and the agent will need to perform the API call daily, but may choose to update the information only if the version changes,
6. Code Table must be carefully analyzed as information returned is critical for successful transaction processing
7. The stateProvinceInfo list returned will grow as MoneyGram adds more states/provinces to countries that have states. Agent application must be able to handle new states/provinces returned for these countries. State fields will always be returned as Optional by the getFieldsForProduct (GFFP) API. If a country is selected from the list that has states available in the codeTable API response, then a state is required. Agent application should change the list of states available depending on the country selected. If the country selected does not have any state list in the codeTable API, then the field should be disabled so data is not collected.
 - a. the fields impacted are: senderState, receiverState, senderPhotoIDState, receiverPhotoIDState, thirdPartyState

codeTableRequest

Field Summary

- [agentAllowedOnly](#)

Field Detail

Tag	Datatype	Information and Rules	Input	Data
agentAllowedOnly	boolean Default value: <code>false</code>	<i>Additional Information</i> 1. If <code>true</code>, the codeTableResponse returns only delivery options available to the agent 2. If <code>false</code>, the codeTableResponse returns all delivery options regardless of whether they are available to the agent MoneyGram recommends to set the <code>agentAllowedOnly</code> flag to <code>true</code> if delivery options are obtained from code tables. Flag can be set to <code>false</code> if delivery options are obtained from <code>feeLookUpResponse</code>	application	optional

codeTableResponse

Field Summary

- version
- stateProvinceInfo
 - countryCode
 - stateProvinceCode
 - stateProvinceName
- countryInfo
 - countryCode
 - countryName
 - countryLegacyCode
 - sendActive
 - receiveActive
 - directedSendCountry
 - mgDirectedSendCountry
 - baseReceiveCurrency
 - isZipCodeRequired
- currencyInfo
 - currencyCode
 - currencyName
 - currencyPrecision
- countryCurrencyInfo
 - countryCode
 - baseCurrency
 - localCurrency
 - receiveCurrency
 - indicativeRateAvailable
 - deliveryOption
 - receiveAgentID
 - receiveAgentAbbreviation
 - mgManaged
 - agentManaged
 - validationExprs
 - checkDigitAlg
- deliveryOptionInfo
 - dssOption
 - deliveryOptionID
 - deliveryOption
 - deliveryOptionName
- rewardsInfo
 - programType
 - name
 - description
- rewardsCardTypeInfo
 - cardType
 - description
- rewardsRegistrationInfo
 - country
 - programType
 - cardType
 - allowPrePrintedCards
 - allowStandardCards

Field Detail

Tag	Datatype	Information and Rules	Display
version	string Maximum length: 20	The code table version returned.	N/A
stateProvinceInfo	complex Maximum number of occurrences: unbounded	<i>Additional Information</i> 1. Currently, data is only returned for Canada, Mexico, and the United States Business rule to code: At present; State is required and must be submitted if Sender/ReceiverCountry & photoIDCountry is equal to Canada, Mexico, the United States, and any country from codeTable API that has state/province lists	N/A
countryCode	string Length: 3	None	N/A
stateProvinceCode	string Length: 2	<i>Additional Information</i> 1. The value is a valid state code to be used whenever a state code is required	N/A
stateProvinceName	string Maximum length: 40	None	N/A
countryInfo	complex Maximum number of occurrences: unbounded	None	N/A
countryCode	string Length: 3	<i>Additional Information</i> 1. The value is a valid country code to be used whenever a country code is required	N/A

Tag	Datatype	Information and Rules	Display
countryName	string Maximum length: 50	None	N/A
countryLegacyCode	string Length: 2	None	N/A
sendActive	boolean	<i>Additional Information</i> 1. If <code>true</code> , agents in the country may perform send transactions 2. If <code>false</code> , agents in the country may not perform send transactions	N/A
receiveActive	boolean	1. If <code>true</code> , agents in the country may perform receive transactions 2. If <code>false</code> , agents in the country may not perform receive transactions 3. List of destination countries must be filtered at the GUI level to include only countries with flag set to <code>true</code>	N/A
directedSendCountry	boolean	<i>Additional Information</i> 1. If <code>true</code> , the country has directed send partners 2. If <code>false</code> , the country does not have directed send partners	N/A
mgDirectedSendCountry	boolean	For internal use only	N/A
baseReceiveCurrency	string Length: 3	The country's default receive currency.	N/A

Tag	Datatype	Information and Rules	Display
isZipCodeRequired	Boolean	Indicates if zip/postal code is required for the country. If true, the zip/postal code must be provided on the transaction (in senderZipCode, receiverZipCode, or thirdPartyZipCode) for the specified country. If false, the zip code field (senderZipCode, receiverZipCode, or thirdPartyZipCode) can be left blank for the specified country.	N/A
currencyInfo	complex Maximum number of occurrences: unbounded	None	N/A
currencyCode	string Length: 3	<i>Additional Information</i> 1. The value is a valid currency code to be used whenever a currency code is required	N/A
currencyName	string Maximum length: 40	None	N/A
currencyPrecision	int	The number of digits used by the currency for fractional amounts.	N/A
countryCurrencyInfo	complex Maximum number of occurrences: unbounded	None	N/A
countryCode	string Length: 3	None	N/A
baseCurrency	string Length: 3	None	N/A
localCurrency	string Length: 3	None	N/A

Tag	Datatype	Information and Rules	Display
receiveCurrency	string Length: 3	None	N/A
indicativeRateAvailable	boolean	<i>Additional Information</i> - If <code>true</code>, the country uses estimated rates if valid rates are not available in the local currency. The exchange rate is not determined at time of the Send transaction, but it's determined at the time of Receiving the transaction - If <code>false</code>, the country does not use estimated rates and exchange rates are guaranteed. Fee information returned during a fee quote will not return any estimated values <code>feeLookup</code> cannot be called with <code>receiveAmount</code>, when the destination country has at least one delivery option with this flag set to true.	N/A
deliveryOption	string	<i>Additional Information</i> If value returned, the country currency information applies only to the specific delivery option	N/A
receiveAgentID	string Length: 8	<i>Additional Information</i> - If value returned, the country currency information applies only to the specific agent ID <code>receiveAgentID</code> is required to complete a Send transaction	N/A
receiveAgentAbbreviation	string Maximum length: 10	The abbreviated form of the agent's name to be displayed on application screen and receipts	N/A
mgManaged	string Maximum length: 20 Valid values: NOT ALLOWED, REQUIRED, optional	For internal use only	N/A

Tag	Datatype	Information and Rules	Display
agentManaged	string Maximum length: 20 Valid values: NOT ALLOWED, REQUIRED, optional	For internal use only	N/A
validationExprs	string Maximum length: 20	Field not currency used	N/A
checkDigitAlg	string Maximum length: 10	<i>Additional Information</i> 1. Field not currently used	N/A
deliveryOptionInfo	complex Maximum number of occurrences: unbounded	None	N/A
dssOption	boolean	<i>Additional Information</i> 1. If <code>true</code> , the delivery option is a directed send option 2. If <code>false</code> , the delivery option is not a directed send option	N/A
deliveryOptionID	int Maximum inclusive value: 999	The delivery option's numeric ID number.	N/A
deliveryOption	string	The delivery option's alphabetic ID. <i>Additional Information</i> 1. The value is a valid delivery option to be used whenever a delivery option is required	N/A
deliveryOptionName	string Maximum length: 40	1. The delivery option's display name. 2. GUI must display the text returned on the <code>deliveryOptionName</code> if <code>deliveryOption</code> is obtained from codeTable API	N/A

Tag	Datatype	Information and Rules	Display
rewardsInfo	complex Maximum number of occurrences: unbounded	Contains information on the MoneyGram Plus program available to the agent's consumers.	N/A
programType	string Maximum length: 3	The program type code of the MoneyGram Plus program available to the agent's consumers. <i>Additional Information</i> 1. Maps to the <code>programType</code> returned in <code>rewardsRegistrationInfo</code>	N/A
name	string Maximum length: 64	The name of the MoneyGram Plus program available to the agent's consumers.	N/A
description	string Maximum length: 40	The description of the MoneyGram Plus program available to the agent's consumers.	N/A
rewardsCardTypeInfo	complex Maximum number of occurrences: unbounded	Contains information on the MoneyGram Plus card available to the agent's consumers.	N/A
cardType	string Maximum length: 4	The MoneyGram Plus card type available to the agent's consumers. <i>Additional Information</i> 1. Maps to the <code>cardType</code> returned in <code>rewardsRegistrationInfo</code>	N/A
description	string Maximum length: 40	The description of the MoneyGram Plus card available to the agent's consumers.	N/A

Tag	Datatype	Information and Rules	Display
rewardsRegistrationInfo	complex Maximum number of occurrences: unbounded	Contains information on MoneyGram Plus program and card types available to the agent's consumers. <i>Additional Information</i> 1. Each occurrence maps the <code>programType</code> to the <code>programType</code> in <code>rewardsInfo</code> 2. Each occurrence maps the <code>cardType</code> to the <code>cardType</code> in <code>rewardsCardTypeInfo</code>	N/A
country	string Length: 3 Must be a valid country code	None	N/A
programType	string Maximum length: 3	This information must be used in Plus Enrollment APIs	N/A
cardType	string Maximum length: 4	This information must be used in Plus Enrollment APIs	N/A
allowPrePrintedCards	boolean	<i>Additional Information</i> 1. If <code>true</code> , the program and card combination allows agents to use pre-printed Plus cards 2. If <code>false</code> , the program and card combination does not allow agents to use pre-printed Plus cards	N/A
allowStandardCards	boolean	<i>Additional Information</i> 1. If <code>true</code> , the program and card combination allows agents to use standard Plus enrollment 2. If <code>false</code> , the program and card combination does not allow agents to use standard Plus enrollment	N/A

3. *countryInfo*

The agent application uses the [countryInfo](#) API to obtain country information required to perform transactions. All information contained in this API is already provided in the CodeTable.

Additional Information

1. The agent application must use the [countryInfo](#) API together with the [currencyInfo](#) and [stateProvinceInfo](#) APIs if CodeTable API is not used
2. Agents must store the information returned in the [countryInfoResponse](#)
3. The agent must create an automated process to obtain updates during non-business hours; the agent must also create a manual process to obtain updates in the event system changes occur during business hours
4. The country information only needs to be updated by one location using the model agent ID; the information may be shared among locations

countryInfoRequest

Field Summary

- [countryFilter](#)

Field Detail

Tag	Datatype	Information and Rules	Input	Data
countryFilter	string Valid values: MG_DIRECTED_SEND, RECEIVE_ACTIVE	<i>Additional Information</i> 1. If value is MG_DIRECTED_SEND, the countryInfoResponse only returns information for countries with directed send/account deposit 2. If value is RECEIVE_ACTIVE, the countryInfoResponse only returns information for countries where agents can perform receive transactions 3. If no value is specified, the countryInfoResponse returns information for all countries	application	optional

countryInfoResponse

Field Summary

- version
- countryInfo
 - countryCode
 - countryName
 - countryLegacyCode
 - sendActive
 - receiveActive
 - directedSendCountry
 - mgDirectedSendCountry
 - baseReceiveCurrency
 - isZipCodeRequired

Field Detail

Tag	Datatype	Information and Rules	Display
version	stringmaximum length: 20		
countryInfo	complex Maximum number of occurrences: unbounded	None	N/A
countryCode	string Length: 3	<i>Additional Information</i> 1. The value is a valid country code to be used whenever a country code is required	N/A
countryName	string Maximum length: 50	None	N/A
countryLegacyCode	string Length: 2	None	N/A
sendActive	boolean	<i>Additional Information</i> 1. If <code>true</code> , agents in the country may perform send transactions 2. If <code>false</code> , agents in the country may not perform send transactions	N/A
receiveActive	boolean	<i>Additional Information</i> 1. If <code>true</code> , agents in the country may perform receive transactions 2. If <code>false</code> , agents in the country may not perform receive transactions	N/A

Tag	Datatype	Information and Rules	Display
directedSendCountry	boolean	<i>Additional Information</i> - If <code>true</code>, the country has directed send partners - If <code>false</code>, the country does not have directed send partners	N/A
mgDirectedSendCountry	boolean	<i>Additional Information</i> - If <code>true</code>, the country has MoneyGram-managed directed send partners - If <code>false</code>, the country does not have MoneyGram-managed directed send partners	N/A
baseReceiveCurrency	string Length: 3 Must be a valid currency code	The country's default receive currency.	N/A
isZipCodeRequired	Boolean	Indicates if zip/postal code is required for the country. If <code>true</code> , the zip/postal code must be provided on the transaction (in <code>senderZipCode</code> , <code>receiverZipCode</code> , or <code>thirdPartyZipCode</code>) for the specified country. If <code>false</code> , the zip code field (<code>senderZipCode</code> , <code>receiverZipCode</code> , or <code>thirdPartyZipCode</code>) can be left blank for the specified country.	N/A

4. *currencyInfo*

The agent application uses the [currencyInfo](#) API to obtain currency information required to perform transactions. All information contained in this API is already provided in the CodeTable.

Additional Information

1. The agent application must use the [currencyInfo](#) API together with the [countryInfo](#) and [stateProvinceInfo](#) APIs if CodeTable API is not used
2. Agents must store the information returned in the [currencyInfoResponse](#)
3. The agent must create an automated process to obtain updates during non-business hours; the agent must also create a manual process to obtain updates in the event system changes occur during business hours
4. The currencyinformation only needs to be updated by one location using the model agent ID; the information may be shared among locations

currencyInfoRequest

currencyInfoResponse

Field Summary

- [currencyInfo](#)
 - [currencyCode](#)
 - [currencyName](#)
 - [currencyPrecision](#)

Field Detail

Tag	Datatype	Information and Rules	Display
currencyInfo	complex Maximum number of occurrences: unbounded	None	N/A
currencyCode	string Length: 3 Must be a valid currency code	None	N/A
currencyName	string Maximum length: 40	None	N/A
currencyPrecision	int	The number of digits used by the currency for fractional amounts.	N/A

5. *stateProvinceInfo*

The agent application uses the [stateProvinceInfo](#) API to obtain state and province information required to perform transactions. All information contained in this API is already provided in the CodeTable.

Additional Information

1. The agent application must use the [stateProvinceInfo](#) API together with the [countryInfo](#) and [currencyInfo](#) APIs if CodeTable API is not used
2. Agents must store the information returned in the [stateProvinceInfoResponse](#)
3. The agent must create an automated process to obtain updates during non-business hours; the agent must also create a manual process to obtain updates in the event system changes occur during business hours
4. The state province information only needs to be updated by one location using the model agent ID; the information may be shared among locations
5. The stateProvinceInfo list returned will grow as MoneyGram adds more states/provinces to countries that have states. Agent application must be able to handle new states/provinces returned for these countries. State fields will always be returned as Optional by the getFieldsForProduct (GFFP) API. If a country is selected from the list that has states available in the stateProvinceInfo API response, then a state is required. Agent application should change the list of states available depending on the country selected. If the country selected does not have any state list in the stateProvinceInfo API, then the field should be disabled so data is not collected.
 - a. the fields impacted are: senderState, receiverState, senderPhotoIDState, receiverPhotoIDState, thirdPartyState

stateProvinceInfoRequest

Field Summary

- [stateProvinceFilter](#)

Field Detail

Tag	Datatype	Information and Rules	Input	Data
stateProvinceFilter	string Valid values: must be a valid country code	<i>Additional Information</i> 1. If country code is provided, the stateProvinceInfoResponse only returns information for states/provinces in that country 2. If no value is specified, the stateProvinceInfoResponse returns state and province information for all available countries	application	optional

stateProvinceInfoResponse

Field Summary

- [version](#)
- stateProvinceInfo
 - countryCode
 - stateProvinceCode
 - stateProvinceName

Field Detail

Tag	Datatype	Information and Rules	Display
version	int	The version number of the state/province information.	N/A
stateProvinceInfo	complex Maximum number of occurrences: unbounded	None	N/A
countryCode	string Length: 3 Must be a valid country code	None	N/A
stateProvinceCode	string Length: 2 Must be a valid state code	None	N/A
stateProvinceName	string Maximum length: 40	None	N/A

6. *translations*

The agent application uses the [translations](#) API to obtain country, currency, delivery option, and other code table information in a variety of languages.

Additional Information

1. Agents must store the information returned in the [translationsResponse](#)
2. The agent must create an automated process to obtain updates during non-business hours; the agent must also create a manual process to obtain updates in the event system changes occur during business hours

translationsRequest

Field Summary

- [languageCodes](#)

Field Detail

Tag	Datatype	Information and Rules	Input	Data																																																									
languageCodes	string	<i>Additional Information – the following languages are supported, but not all translations are returned in all the languages listed.</i> 1. Can use 2 letter or 3 letter valid language code 2. Field may contain more than one value as a comma-separated list 3. If no value is included, the translationsResponse returns translations for all languages 4. Available language list <table><tr><th>2 letter code</th><th>3 letter code</th><th>language</th></tr><tr><td>ar</td><td>ara</td><td>Arabic</td></tr><tr><td>bg</td><td>bul</td><td>Bulgarian</td></tr><tr><td>zh</td><td>zho</td><td>Chinese</td></tr><tr><td>cs</td><td>ces</td><td>Czech</td></tr><tr><td>da</td><td>dan</td><td>Danish</td></tr><tr><td>nl</td><td>nld</td><td>Dutch</td></tr><tr><td>en</td><td>eng</td><td>English</td></tr><tr><td>fi</td><td>fin</td><td>Finnish</td></tr><tr><td>fr</td><td>fra</td><td>French</td></tr><tr><td>de</td><td>deu</td><td>German</td></tr><tr><td>el</td><td>ell</td><td>Greek, Modern</td></tr><tr><td>hu</td><td>hun</td><td>Hungarian</td></tr><tr><td>it</td><td>ita</td><td>Italian</td></tr><tr><td>no</td><td>nor</td><td>Norwegian</td></tr><tr><td>pl</td><td>pol</td><td>Polish</td></tr><tr><td>pt</td><td>por</td><td>Portuguese</td></tr><tr><td>ro</td><td>ron</td><td>Romanian</td></tr><tr><td>ru</td><td>rus</td><td>Russian</td></tr></table>	2 letter code	3 letter code	language	ar	ara	Arabic	bg	bul	Bulgarian	zh	zho	Chinese	cs	ces	Czech	da	dan	Danish	nl	nld	Dutch	en	eng	English	fi	fin	Finnish	fr	fra	French	de	deu	German	el	ell	Greek, Modern	hu	hun	Hungarian	it	ita	Italian	no	nor	Norwegian	pl	pol	Polish	pt	por	Portuguese	ro	ron	Romanian	ru	rus	Russian	application	optional
2 letter code	3 letter code	language																																																											
ar	ara	Arabic																																																											
bg	bul	Bulgarian																																																											
zh	zho	Chinese																																																											
cs	ces	Czech																																																											
da	dan	Danish																																																											
nl	nld	Dutch																																																											
en	eng	English																																																											
fi	fin	Finnish																																																											
fr	fra	French																																																											
de	deu	German																																																											
el	ell	Greek, Modern																																																											
hu	hun	Hungarian																																																											
it	ita	Italian																																																											
no	nor	Norwegian																																																											
pl	pol	Polish																																																											
pt	por	Portuguese																																																											
ro	ron	Romanian																																																											
ru	rus	Russian																																																											

		sk	slk	Slovak		
		es	spa	Spanish		
		sv	swe	Swedish		
		tl	tgl	Tagalog		
		th	tha	Thai		
		tr	tur	Turkish		

translationsResponse

Field Summary

- translationsVersion
- countryTranslations
 - countryCode
 - languageCode
 - longLanguageCode
 - displayName
- deliveryOptionTranslations
 - deliveryOption
 - languageCode
 - longLanguageCode
 - displayName
- currencyTranslations
 - currencyCode
 - languageCode
 - longLanguageCode
 - displayName
- fqdoTextTranslations
 - receiveCountry
 - deliveryOption
 - receiveAgentID
 - receiveCurrency
 - languageCode
 - longLanguageCode
 - registrationAuthorizationText
 - speedOfDeliveryText
- industryTranslations
 - industryId
 - displayName
 - languageCode
 - longLanguageCode

Field Detail

Tag	Datatype	Information and Rules	Display
translationsVersion	string	The translation version returned.	N/A
countryTranslations	complex Maximum number of occurrences: unbounded	None	N/A
countryCode	string Length: 3 Must be a valid country code	The country code of the country to which the translation applies.	N/A
languageCode	string	The language into which the country name has been translated.	N/A
longLanguageCode	string Maximum length: 3	The 2 letter or 3 letter valid language code	N/A
displayName	string	The translated name of the country. <i>Field Rules</i> 1. Up to 50 characters if language is English 2. Up to 2000 characters if language is not English	N/A
deliveryOptionTranslations	complex Maximum number of occurrences: unbounded	None	N/A
deliveryOption	string Must be a valid delivery option	The delivery option to which the translation applies.	N/A

Tag	Datatype	Information and Rules	Display
languageCode	string	The language into which the name of the delivery option has been translated.	N/A
longLanguageCode	string Maximum length: 3	The 3 letter language code	N/A
displayName	string	The translated name of the delivery option.	N/A
currencyTranslations	complex Maximum number of occurrences: unbounded	None	N/A
currencyCode	string Length: 3 Must be a valid currency code	The currency code of the currency to which the translation applies.	N/A
languageCode	string	The language into which the name of the currency has been translated.	N/A
longLanguageCode	string Maximum length: 3	None	N/A
displayName	string	The translated name of the currency.	N/A
fqdoTextTranslations	complex Maximum number of occurrences: unbounded	None	N/A
receiveCountry	string Length: 3 Must be a valid currency code	None	N/A

Tag	Datatype	Information and Rules	Display
deliveryOption	string Must be a valid delivery option	None	N/A
receiveAgentID	string Length: 8	None	N/A
receiveCurrency	string Length: 3 Must be a valid currency code	None	N/A
languageCode	string	The language into which the FQDO text has been translated.	N/A
longLanguageCode	string Maximum length: 3	None	N/A
registrationAuthorizationText	string	The FQDO's translated registration authorization text.	N/A
speedOfDeliveryText	string	The FQDO's translated speed of delivery text.	N/A
industryTranslations	complex Maximum number of occurrences: unbounded	None	N/A
industryID	string Maximum length: 6	The industry to which the translation applies.	N/A
displayName	string	The translated name of the industry.	N/A
languageCode	string	The language into which the industry name has been translated.	N/A
longLanguageCode	string Maximum length: 3	The 3 letter language code	N/A

7. *doddFrankStateRegulatorInfo*

The agent application uses the [doddFrankStateRegulatorInfo](#) API to obtain state regulator contact information to print on pre-payment disclosure and receipts. This is required for US and US territories only. Please see Receipt Module for receipt details.

doddFrankStateRegulatorInfoRequest

Field Summary

- [dfJurisdiction](#)

Field Detail

Tag	Datatype	Information and Rules	Input	Data
dfJurisdiction	string Length: 3	The US state or territory for which to return regulator information. US states are 2 letter state codes. US territories are 3 letter territory/country code. <i>Additional Information</i> 1. If no value is provided, the response returns information for all states and territories 2. State regulatory information will be returned for US states and US territories only.	application	

doddFrankStateRegulatorInfoResponse

Field Summary

- version
- stateRegulatorInfo
 - dfJurisdiction
 - stateRegulatorURL
 - stateRegulatorPhone
 - stateRegulatorName
 - longLanguageCode
 - textTranslation

Field Detail

Tag	Datatype	Information and Rules	Display
version	int	The version number of the state regulator information.	N/A
stateRegulatorInfo	complex Maximum number of occurrences: unbounded	None	N/A
dfJurisdiction	string	The code for the US state or US territory	N/A
stateRegulatorURL	string Maximum length: 2000	The state regulator's Web site address.	N/A
stateRegulatorPhone	string Maximum length: 100	The state regulator's telephone number.	N/A
stateRegulatorName	complex Maximum number of occurrences: unbounded	None	N/A
longLanguageCode	string Length: 3	The language state regulator's name.	required
textTranslation	string Maximum length: 2000	The translated regulator's name. The Dodd-Frank Remittance Transfer Rule requires that the data returned in the <code>textTranslation</code> field for a particular state or territory must be printed on the Money Transfer receipts. Reason: To be in compliance with the Dodd-Frank Wall Street Reform and Consumer Protection Act	required

8. variableReceiptInfo

The agent application uses the [variableReceiptInfo](#) API to obtain variable receipt information to print on redirect or customer terms acceptance on Agent Copy receipt during a Money transfer transaction.

MoneyGram uses Unicode to encode character strings that need to be displayed on screens and/or receipts. Unicode is a standard which defines a universal character set. In XML, representations of Unicode are: `Ӓ` or `􏿽`

For example,

ó is Unicode (U+00F3) which is decimal 243. This can be represented in XML as either `ó` or `ó`

AgentConnect clients will need to use the appropriate parsing algorithms to convert from the XML Unicode representation to the encoding needed for their own display devices and printers.

See <http://www.w3.org/TR/unicode-xml/> for more information on the XML Unicode standard and <http://www.unicode.org/charts/> for information on Unicode characters themselves.

variableReceiptInfoRequest

Field Summary

- [longLanguageCode](#)

Field Detail

Tag	Datatype	Information and Rules	Input	Data
longLanguageCode	string Length: 3 Must be a valid 3-letter language code	The language in which receipt information to be returned. <i>Additional Information</i> 1. Valid values: spa: Spanish, eng:English	Application/ GUI	required

variableReceiptInfoResponse

Field Summary

- version
- receiptText
 - receiptTextType
 - textTranslation
 - longLanguageCode
 - textTranslation

Field Detail

Tag	Datatype	Information and Rules	Display
version	int	The version number of the variable receipt information.	Optional
receiptText	complex Maximum number of occurrences: unbounded	None	Optional
receiptTextType	string	Type of receipt such as termsAcceptanceSend, redirectInfoReceive, etc.	Optional
textTranslation		To be in compliance with the Dodd-Frank Wall Street Reform and Consumer Protection Act	Optional
longLanguageCode	string length: 3	The 3 letter language code in which the regulatory text is returned.	Optional
textTranslation	string Maximum length: 2000	The translated verbiage to be printed on the receipt. Translated text in the language specified in the long language code. The text will be printed on Money Transfer receipts in the case of redirects as part of regulatory information.	Required

9. industry

The agent application uses the [industry](#) API to obtain industry ID and names information for bill payment.

industryRequest

Field Summary

- (None)

Field Detail

Tag	Datatype	Information and Rules	Input	Data
(None)	N/A	None	N/A	

industryResponse

Field Summary

- industryInfoList
 - industryID
 - industryName

Field Detail

Tag	Datatype	Information and Rules	Display
<u>industryInfoList</u>	complex Maximum number of occurrences: unbounded	None	N/A
industryID	String 6 characters	Generally the three letter code of Industries are returned.	N/A
industryName	string	Name of the industry, such as GAS, ELECTRIC, MORTGAGE etc.	N/A

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